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Why Should Surface Runoff Analysis Become a Permanent Part of the Climate Change Adaptation Strategy for Medium-Sized Cities?

Abstract

Climate fluctuations can result in long periods of drought interspersed with short but rapid and intense rainfall events. This feasibility study includes verification of the planning documents for a medium-sized city of Katrineholm (including historical and problematic sites) and Land-use-land-cover modeling (LULC). Aim was to identify the best Nature-based Solution (NBS) that could be incorporated into the local adaptation plan for reducing surface runoff and improving retention. In SCALGO, the data for modeling of surface runoff are based on topography, LULC and soil data. Smart surface water management on a catchment level was performed using 'flash flood' method with disabled sewerage system for Low-Impact Development (LID), Bio-density and Urban Greenery (UG). Each of the models improves the situation. The most effective NBS was UG within catchments of historic city center and industrial urban typologies (benefits between +43% and +225%), surpassing LID (+81 to +129%), while Bio-density showed minor alterations (+9 to 21%). The change in LULC affected up to 34% of residential, up to 62% of industrial and up to 48% for downtown. Research findings can be interpreted in terms of the quantitative benefits of reduced surface runoff and in urban planning (reaching for climate resiliency) and protection of historical parts.

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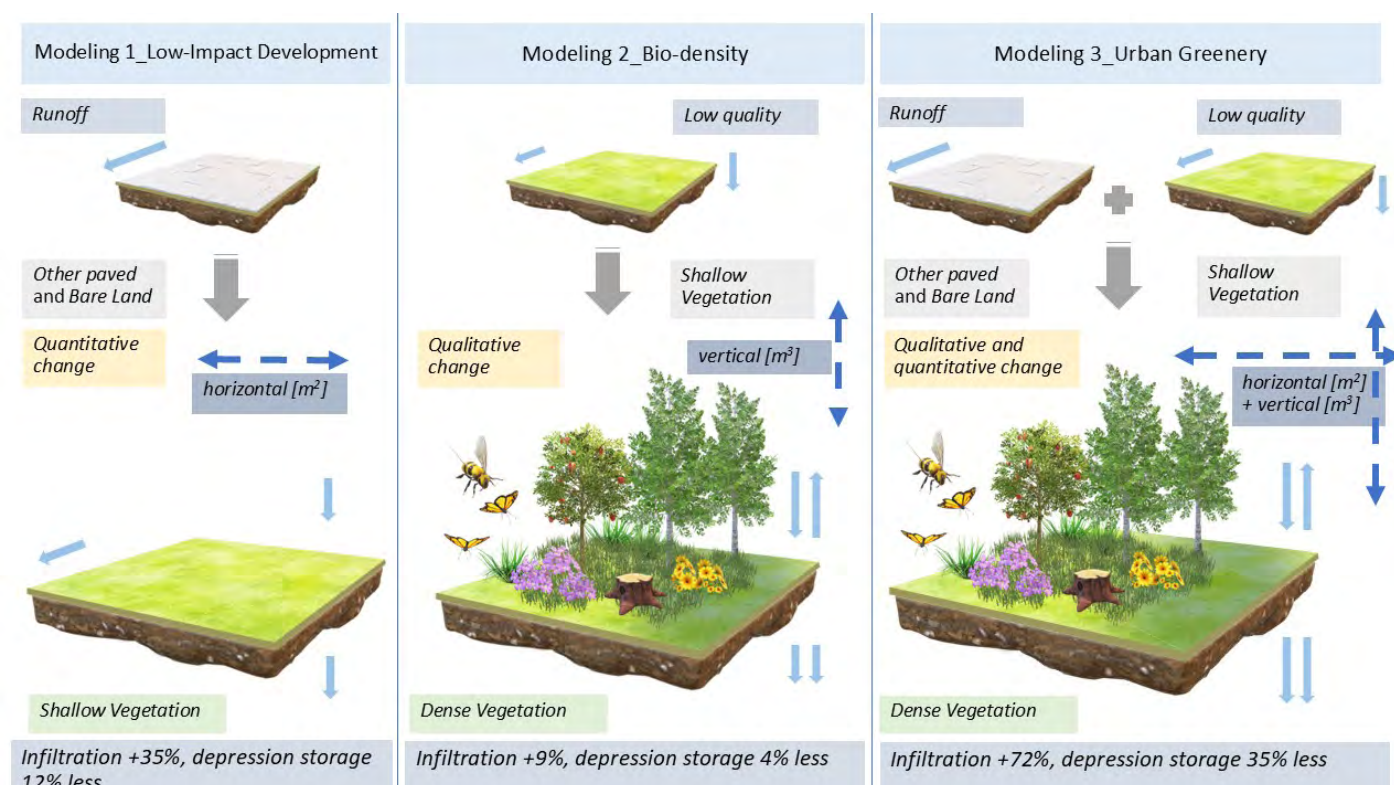
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Graphical abstract



1 Introduction

Being a part of landscape and urban planning, nature-based solutions (NBS) and blue-green infrastructures (BGI) are comprehensive, most sustainable and leading remedies for climate change adaptation (Castellar et al., 2024; Suleiman, 2021; Reinwald et al., 2024; Longato et al., 2024; Mickovski et al., 2024; Aghaloo et al., 2024). The use of NBS has a positive impact on both mitigation and adaptation to counteract climate variability on a relatively low-cost basis, beneficial both to human and natural resources (Seddon et al., 2020; UN, 2015). BGI, being interconnected natural, semi-natural systems (Wu et al., 2024; Aghaloo et al., 2024) and constructed ecosystems (Aghaloo et al., 2024), can, among other objectives, regulate surface runoff (Paggioli et al., 2024; Staccione et al., 2024) or help reduce stormwater runoff pollution (Cavadini et al., 2024). All this considered, a synergy of rainwater management systems can be an effective way of mitigating droughts and floods (Wu et al., 2024; Li et al., 2024; Staccione et al., 2024; Aghaloo et al., 2024; Burns et al., 2012).

Climate fluctuations, intensifying for the last 30 years, can result in long periods of drought interspersed with short but rapid and intense rainfall events (IPCC, 2019; IPCC, 2023; Muller et al., 2015). Those changes affect cities, likewise, threatening with cumulative surface water both ways: directly (threat has arisen locally, and is a result of down-pour within city borders), or indirectly (threat radiates from neighboring areas). A promising solution to tackle such risks is a systemic surface water management, considered at a catchment level (Iwanicki, 2007; Banasik et al., 2014; Sveriges Riksdag, 2018). Spread across all continents and climate zones (Li et al., 2024), various tedious water disruptions mainly affect urbanized and constantly developing areas (Aghaloo et al., 2024) noted primarily for the risk of financial burdens (Hallegatte et al., 2013), loss of human health, or lives (Luo et al., 2024). Increasing pressure of urbanization and concentrated development are leading to urban sealing, resulting in accelerated surface runoff, reduction in green spaces, and creation of rainwater accumulation areas (Jacobson, 2011). In addition to this, compounded irregularities in water runoff can be triggered by a convergence of

climatic fluctuations and land-cover-land-use modifications (Huq & Abdul-Aziz, 2021). Limited capacity of drainage systems made of pipes and concrete structures has proved insufficient to cope with severe weather conditions (Soni et al., 2025; Assamuddin et al., 2025). This evolving situation highlights a need for more sophisticated (smarter) water management in urban settings.

Land-use planning employs environmental engineering measures to reduce climate change-related risks. Numerous urban planning concepts promote a variety of approaches, many of which may be based on the same NBS tools. Green Infrastructure (GI) is being considered as a solution to minimize the risks of excessive surface runoff and flooding (Liu et al., 2014; Webber et al., 2019; Assamuddin et al., 2025). However, single and centralized GI intervention's capacity was proved limited regarding rainwater capture and risks of flooding. An important aspect highlighted by researchers is raising public awareness of the potential to mitigate the effects of climate change through the implementation of GI. The Garden city concept (GCC; Howard, 1898) was not originally conceived as a solution to improve stormwater management, including surface runoff; however, today we can also recognize this potential in the urban developments created on the basis of this approach. In contemporary planning solutions, GCC is viewed by urban planners as elements of ecological corridors and landscapes, providing ecosystem services, serving as means of transport (e.g. canals) and flood prevention measures (ZeXina et al., 2011).

Sustainable Urban Drainage System (SUDS, developed in UK, Fletcher et al., 2015) and LID (Low Impact Development; origins in North America; Javan et al., 2025; Sehwat & Shekhar, 2025; Ahiablame & Shaky, 2016; Soltaninia & Eskandaripour, 2026) are a network of engineered "green" initiatives which can appear as roof and rain gardens, porous pavements (Tang et al., 2021) and underground water tanks/collectors (Ruiz-Perez et al., 2022). Being a representative for both systems, permeable pavements show the most effective outcomes in runoff reduction and flood protection in long-term periods (e.g. 50-year return period) and large-scale implementations (Essamlali et al., 2024). SUDS were also modeled using the water footprint approach for green infrastruc-

ture (Hoekstra et al., 2011) within the water life cycle, which is based on three components: blue, green and grey infrastructure (Ruiz-Perez et al., 2022). The data were grouped into input data comprising water used for irrigation (input amount of water), greywater system (output amount of water) and water used in biological processes (rain, evaporation and plant transpiration). The study utilized the Curve Number for permeable (infiltration) and impermeable areas (drainage to the sewer system) to calculate water volumes for an existing street renovation project in Spain. Urban Green Infrastructures (UG or UGI) are primarily used to promote wellbeing and reduce urban heat; their use for stormwater and flood management is very rarely described. Their preventive application in minimizing flood risks and their impacts is a relatively new but promising approach at the catchment scale (Uribe et al., 2022). The Sponge City (SPC) concept, developed in China in 2014 (Li et al., 2016; Luo et al., 2024; Chaturvedi et al., 2025), stems largely from earlier systems implemented in China, namely LID, the Sustainable Urban Drainage System (SUDS), Low Impact Urban Design and Development (LIUDD) and Water Sensitive Urban Design (WSUD). Bio-density is a concept used at the intersection of urban planning, landscape ecology and spatial planning. It is a new theory, combining the desire to design a compact city with the preservation of dense vegetation and biodiversity. One of the key elements is the proportion of biologically active areas, the spatial distribution of individual green spaces and their connectivity. Biological density refers to the concentration of plant biomass (Yu et al., 2024).

Surface runoff is the most widespread risk of climate change which requires consideration as a feasibility study to any adaptation strategy within an entire catchment area and at every administrative level. The optimized algorithms can now be applied as part of an initial, rapid simulation, even with limited input data identifying areas within the urban fabric where BGI could help to solve the problem of flooding, even without the support of drainage systems (Li et al., 2024). The priority should be given to areas where the type and quality of land cover determine the effectiveness and scale of Nature-Based Solutions (NBS) in protecting cultural heritage, human lives, and urban infrastructure (Staccione et

al., 2024; Wang et al., 2024). Neither references nor studies in literature addressing this relationship have been found. Research into application of sustainable methods of NBS (Soni et al., 2025) in small and medium-sized towns currently focuses on their incorporation into a spatial planning process: recognizing and implementing their benefits (Giedych et al., 2026), morphology of BGI elements within spatial settings based on urban planning literature (Cieszewska et al., 2024) or identifying barriers to their deployment (Legutko-Kobus et al., 2025). With this in mind, research into urban green spaces (BGI) in smaller towns focuses primarily on the spatial context (dimensions in m²) relating more to the search for ways to improve urban planning (future developments), but without taking water management into account.

Hence, our research is aimed to identify practical tools for assessing the scale and location of surface runoff hazards in medium-sized cities and for managing rainwater as a part of policy development, pointing the way towards improving existing urban structures. To assess the response of individual land-use categories, a tool for analyzing flash floods can be used, which enables the modeling of surface runoff on a catchment-wide scale (Banasik et al., 2014; Daigneault et al., 2016; Paggioli et al., 2024; Li et al., 2024). The main parameters that influence the volume of runoff can be identified as 1) intensity of rainfall (amount and time), 2) size of the catchment, 3) Land use/Land cover (LULC), 4) urban typology (with the level of imperviousness), 5) soil type (infiltration capabilities), 6) location in the basin, 7) efficiency/capacity of the sewerage system, 8) topography, and others (Banasik et al., 2014; Paggioli et al., 2024; Li et al., 2024; Seddon et al., 2020; Daigneault et al., 2016; Walsh et al., 2015; Burns et al., 2012; Balstrøm, 2024; Mickovski et al., 2024; Naz et al., 2015; Kumar et al., 2024; Scuito et al., 2025; Wu et al., 2024).

In this article, we examine an extent to which NBS can be incorporated in a medium-sized city, while also evaluating the results of managing excessive surface runoff through alterations in land-use. This will make it possible to determine to what extent NBS solutions can be incorporated into a climate change adaptation strategy as a planning tool, a topic that has not been thoroughly explored in the

literature. Using a case study – a specific urban ecosystem and a local authority/city level (Morris et al., 2018; Standish et al., 2014; Walsh et al., 2015; Lebrun et al., 2026) – we propose a simple, novel and systematic methodological approach. This feasibility study for an adaptation action plan for medium-sized cities, with limited access to input data, is based on a two-stage research project and three types of spatial-engineering approaches.

The research objectives of this study are to identify the most effective NBS systems in terms of infiltration capacity, with a view to reducing surface runoff. The research hypotheses are that 1) it is possible to implement NBS within existing urban structures, including those of high cultural (historical) value in small and medium-sized town centers in a way that would reduce surface runoff and improve infiltration capacity; 2) it is possible to develop problematic spaces adjacent to historic city centers in a way that would protect them from the negative effects of excessive surface runoff; 3) the modeling on permeabilization of impervious surfaces in city centers/built-up areas (even in small and medium-sized towns) can yield tangible benefits in terms of reducing surface runoff and increasing infiltration.

2 Materials and methods

This research was divided into two stages where various examination methods were implemented (Fig. 1, Table 1.). This is an exercise that has a universal application.

Phase 1 “Horizontal”/ Identification: planning document analysis and search for problem areas regarding rainwater retention (accumulation sites, retention potential of land cover types) in software for early detection and simulation of surface runoff effects (blue spots);

Phase 2.1 “Vertical”/ Modeling: application of measurable and attainable NBS tools, which can introduce qualitative and quantitative alterations into public space on exemplary parts of a medium-sized city: (1) low-impact development (LID) – unsealing, understood as restoring permeability; (2) Bio-density (BD) – strengthening biodiversity through a use of a multi-layered vegetation cover structure; high-

quality, dense green spaces; (3) Urban greenery (UG) – maximal synergies possible based on incorporation of diverse green spaces and dense vegetation (vegetation as a coherent, continuous and multifunctional natural structure integrated into urban fabric). Measurements indicate the amount of water infiltrated or drained and stored in depressions.

Phase 2.2: “Consensus”/ NBS elements specification: indicating the most excellent effectiveness of selected BGI tools in delaying surface runoff, increasing retention and considering a contribution of stormwater drainage in urban areas.

Being of a general nature, a case study for the city of Katrineholm (central Sweden) serves to conduct an investigation of the best practice solutions from Low-impact development (LID), Bio-density (BD) and Urban Greenery (UG) within a described scope. The selection of the case study site was determined by the topography of terrain and water absorption capacity of the soils in the selected areas and was dictated by the availability of planning documents and thematic maps.

Materials used for the first phase included a set of **planning documents** from Katrineholm (Översikplan,

2014; Klimatstrategi för Katrineholms kommun, 2005; Miljöpolicy, 1999; Grönplan, 2018). This municipality is a specific place with an existing Adaptation Plan to Climate Change, which is a significant strategy that can be seen as a pioneering one, since only few medium-sized European cities have already developed one (and are trying to implement it). Using a qualitative method, the content of those records was verified in terms of development’s spatial distribution, scale of remediation, and methods of how excessive surface runoff is being counteracted. In this regard, **meteorological data** were derived from weather stations for Norrköping for a period between 2001 and 2025 (Meteomanz, n.d.; Wunderground, n.d.), Katrineholm (timespan of 2020-2024) (Wunderground, n.d.). Additional data on annual averages for Sweden (2021-2025; SMHI, 2025) served to identify possible trends in precipitation in Norrköping relative to the rest of the country, in order to identify fluctuations, which could indicate problems with an excess runoff and possible trends of climate regime shifts (temperature, precipitation). A descriptive method was used to summarize the results from two neighboring cities lying in the same regional catchment and for Sweden. A number of inhabitants and trends in the city’s

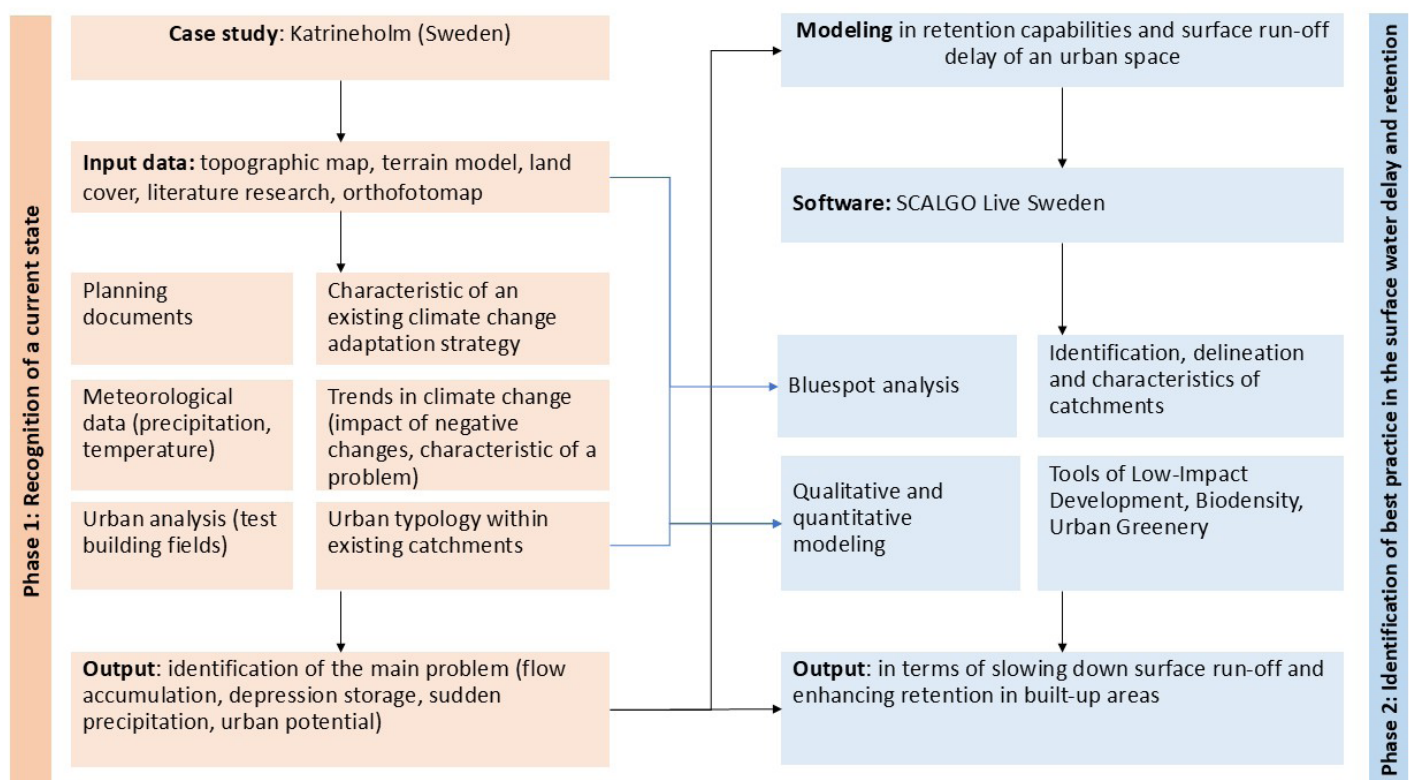


Figure 1. Research methodology

Table 1. Materials and methods for a two-stage methodology. Own elaboration.

Phase 1. Input data	Derived from...	Characteristics	Goal	Methods
Number of inhabitants	Statistic Sweden	Own elaboration	Identification of the size of the locality (a medium-sized town) and trends in change	Identification
Precipitation, temperature	Meteomanz.com, Wunderground.com	Own elaboration	Analysis of figures and trends	Identification, statistics
Land cover	SCALGO Live Sweden/ Lantmäteriet.se	Resolution 25cm, AI based	Identification of the current situation, urban typology	Identification, calibration
Orthophoto map	SCALGO Live Sweden/ Lantmäteriet.se	Resolution from 16cm to 50cm	Identification of the current situation	Identification, calibration
Terrain model DEM	SCALGO Live Sweden/ Lantmäteriet.se	1x1m raster DEM	Identification of the current situation	Identification, calibration
Topsoil map	SCALGO Live Sweden/ Sveriges Geologiska Undersökning GE. Jordarter https://www.sgu.se/	From 1:25 000-1:100 000 to 1:1 million	Identification of the current situation	Identification, calibration
A set of planning documents as an existing climate change adaptation strategy	Katrineholm.se	Own elaboration	Identification of the current situation	Identification, validation
Rain event	SCALGO Live Sweden/ Swedish Meteorological and Hydrological Institute (SMHI)	50mm/60min	Identification of the current situation	Identification, calibration
Phase 2. Modeling	Conducted in...	Characteristics	Goal	Methods
Land cover	SCALGO Live Sweden	Own settings/ elaboration	Verification: (1) low-impact development (LID) – unsealing, understood as restoring permeability; (2) Bio-density – multi-layered vegetation cover structure; (3) Urban greenery (UG)	Modeling – changing parameters infiltrated or drained, statistics
Rain event	Dahlström, 1979; Dahlström, 2010; SCALGO Live Sweden	100mm/60min		Modeling, calibration

growth were identified on the basis of **population figures** for the municipality and commune of Katrineholm. An analysis of rainfall for the two months with the lowest and highest average rainfall (April and July) indicate an increased average of monthly rainfall compared to rainfall across the whole of Sweden (SF1. in additional materials). As shown in Table 1, SCALGO Live Sweden was helpful to obtain and calibrate essential, up-to-date georeferenced data (Bjork et al., 2024; Larsen et al., 2021; Eggert et al., 2025; Mantilla et al., 2025), such as **orthophoto maps, digital elevation models and topsoil maps**. This software also provides an AI-based land cover layer which, regardless of the zoom level, depicts an actual land cover, expressed in square meters and percentage of the catchment's total. SCALGO's classification for Land cover layers is as follows: *Building, Paved road, Unpaved road, Other paved, Dense vegetation, Shallow vegetation, Bare land, Water*. Selected control and modeling groups of watersheds within city core consist altogether of 47.59% natural areas (see also Table 2).

Using SCALGO Live Sweden, a 'flash flood' analysis (SCALGO Live Sweden; Kotyra & Chabudziński, 2023; Bjork et al., 2024) was conducted designating catchments of the city center as blue spots formed during a single extreme rainfall event with an intensity of 100 mm and duration of 60 min or 5-year return period, according to the common dimensioning practice in Sweden (Dahlström, 1979; Dahlström, 2010; SCALGO Live Sweden). Flash Flood Mapping is used to analyze surface runoff during a specific rainfall event for a selected location. The program generates runoff lines for the catchment area and precipitation accumulation using SCS-CN method/ Horton mechanism. This analysis is directly linked to the elevation model, land cover type and soil type (infiltration capacity). This software can also be used to construct cross-sections of terrain (Roth et al., 2024), create a novel functionality along with other tools for spatial analyses of water management (Wałęga et al., 2024), visualize the results of urban flood risk events and flow accumulation within a watershed (Wałęga et al., 2024; Borowska-Stefańska et al., 2024; Grzybowski et al., 2023; Odgaard et al., 2017), and many more.

Hence, locations of excessive water accumulation and runoff directions within Katrineholm inner city area and all 6 adjacent catchments were delineated. Based on/within those watersheds, an urban typology analysis was conducted to identify types of development fields and urban densities with a proportion of urban green spaces (e.g. industrial or residential areas), as well as their spatial distribution. This phase made it feasible to recognize challenging sites, interpret collected data and identify possible, preliminary, and remedial actions. The first stage of the stormwater management analysis of our case study consisted of a preliminary spatial and urban planning evaluation covering land-cover-land-use, as well as the size and distribution of built-up and green areas. The origins of a variety of buildings and urban complexes can be traced back to the founding plans as a garden city in 1863 (historical area). Additionally, a zone of the city center (historical area) was manually delineated. Further investigation of the problematic areas helped to identify the accumulation sites and surface runoff directions.

Modeling of specific land cover groups is the second phase of this research. In the course of the simulation, we decided to disable the function of rainwater drainage via the sewerage system – an additional feature that can be enabled or disabled in SCALGO – in order to compare the “natural” effectiveness of individual interventions. This allows us to assess the initial impact of those solutions using an example of a specific, laboratory-like ecosystem of a small town. To investigate which alterations (qualitative and/or quantitative) would give the best results in terms of increased retention and delayed runoff, we chose three systems of Nature-based solutions: LID simulation is based on conversion within land cover types from *Other paved* and *Bare Land* to *Shallow Vegetation* (it imitates unsealing and restoration of an infiltration capacity of previously impermeable surfaces); Bio-density: change from *Shallow Vegetation* to *Dense Vegetation*. Urban Greenery transforms *Other paved*, *Bare Land* and *Shallow Vegetation* into *Dense Vegetation* – both qualitative and quantitative change of land cover (and retention capacity) co-occurred here. Furthermore, a topographic map/terrain model (restricted access to data from Geotorget Lantmäteriet, n.d.) was used to investigate a land absorption potential for rainwater in local catchments

of Katrineholm town center. The aim is to be able to incorporate directly into urban planning system and adaptation strategy tailored recommendations for the city of interest/studied.

Non-parametric Friedman tests were applied to assess different results across various scenarios, treating catchments as repeated units. A limitation of this study is that it is a case study and does not involve a random selection of the study area. The values analyzed are not field measurements subject to random variation, but results generated by a deterministic model. Consequently, this study should be regarded as preliminary and for information purposes, and should be combined with other analyses concerning other examples of small and medium-sized towns or urban areas to enable generalizations to be drawn. All statistical analyses and graphical visualizations were performed in R (version 4.5.2 (2025-10-31)) using the RStudio environment and the ggplot2 package.

3 Results

3.1 Phase one – recognition of a current state

Based on statistical data (Statistics Sweden, 2022), Katrineholm can be classified as a small-medium-sized city (34,765 residents) with a significant growth trend for Europe over a 20-year period, at a rate of 7.2% for the urban area and 11.2% for the administrative area. According to all IPCC climate scenarios for this region, vegetation season is lengthening, and number of days with low soil moisture increasing (Sverigesnatur, n.d.). The amount of water runoff (annual mean) is expected to grow by +7% by 2040 (Climate Information, 2024).

In addition to this, individual rain events have intensified since 2012. Fluctuation and accumulation occur similarly for both cities, and 2023 was extreme regarding the distribution of monthly rainfall totals. In Katrineholm, totals spread between 3.80mm and 39.75mm for February and June while being surpassed by 111.04mm and 159.03mm in July and August, respectively. Moreover, despite an overall decrease in annual precipitation indicating progressive drought, rainfall is becoming more extreme, with intense downpours following dry periods, which

may contribute to an increase in surface runoff. Meanwhile, the distribution of monthly temperatures (lowest, average, and highest) remains stable without significant fluctuations (SF2. in additional materials).

An analysis of the provisions of the spatial development plans was also carried out to determine whether local planning had taken the issue of stormwater management into account. Analyzing four primary planning documents of Katrineholm, such as General Plan 2030 for the city and rural areas (Översikplan, 2014), Climate strategy (Klimatstrategi för Katrineholms kommun, 2005), Environmental strategy (Miljöpolicy, 1999) and Greenery plan (Grönplan, 2018), it can be stated that only the General and Greenery Plans include, also when by and large, provisions for adaptive measures within rainwater management. General plan (from 2014) informs that (1) guidelines for stormwater management in Katrineholm municipality for rural areas were under development; (2) stormwater management was to be introduced at an early planning stage and treated as a resource in urban planning; (3) delayed runoff of rainwater or its local disposal was to be prioritized. Greenery plan contains provisions that the emphasis in urban planning was to be put on multifunctional greenery that can (1) provide clean water, (2) delay stormwater and (3) (while superficial) support biodiversity,

(4) reduce risk of flooding, or (5) avoid erosion damage. The planning documents urged to assess the possibilities of local stormwater management in the EASTERN part of the city.

Using flash flood method, six interconnected catchments with a total area of 4.09 km² were delineated and classified into main land use types. Catchments No. 1, 2, 4 occupy dense mixed use and dense residential blocks with a low share of greenery parts, catchments No. 3, 5 – areas with dominant industrial usage, No. 6 – areas with dominant single and multi-family houses with a significant share of green spaces (Tab. 2 and Fig. 2.) This information was lacking in planning documents from Katrineholm (Fig. 3). Direction of surface runoff within a precinct follows the east-western topography, threatening the historical parts of the city.

When analyzing catchments in terms of land cover in the baseline condition (here: m0, with the sewerage system disabled; Tab.2), it should be emphasized that within the scope a relatively small proportion of areas with *Shallow vegetation* can be found. A high percentage of sealed surfaces (over 72%) occurs in catchments No. 1, 2, 3 and 4 – in central parts of the inner city, the lowest in No. 6 (46.5%) and 5 (50.4%) – in eastern parts (see also ST1. In additional materials).

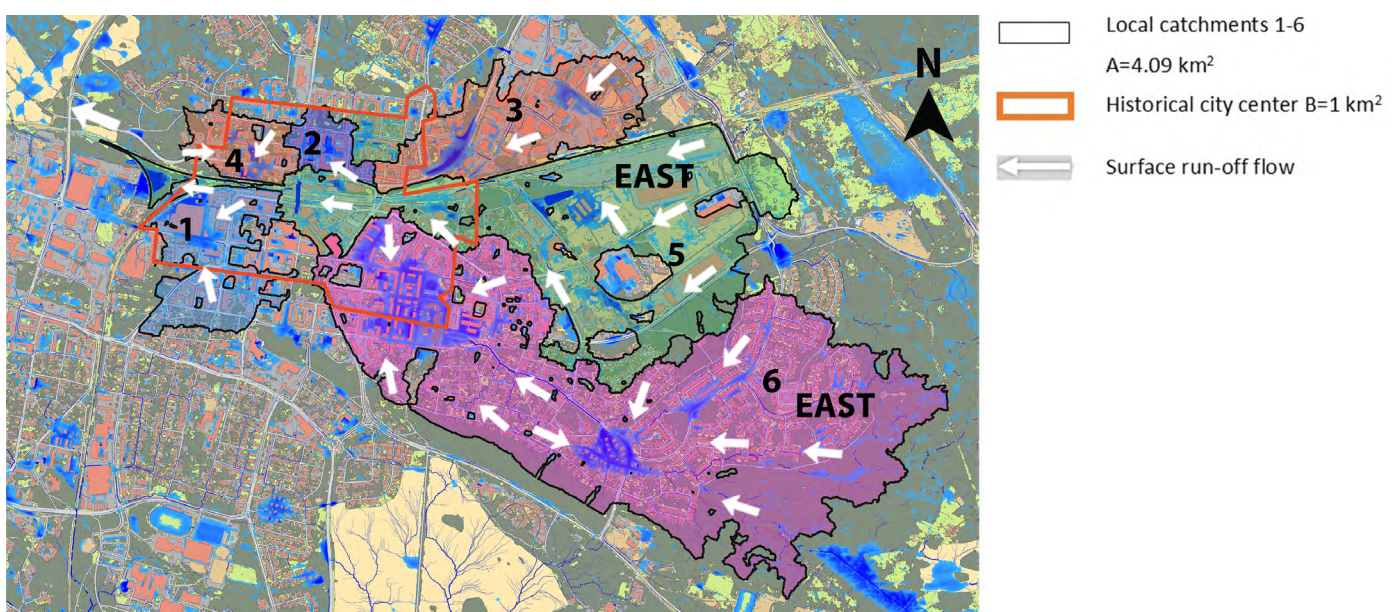


Figure 2. Catchments within the city core of Katrineholm (in the background land use and flow accumulation layers). A zone of the city center (historical area) of 1 km² was manually delineated. Own elaboration based on SCALGO Live Sweden.

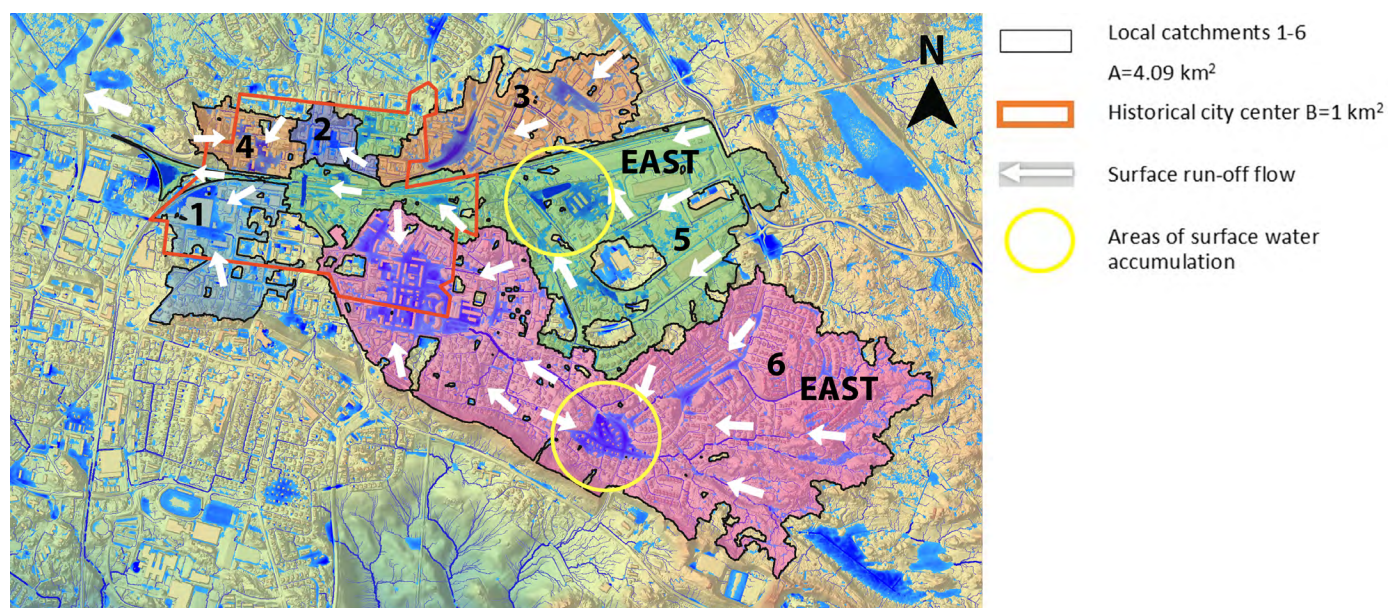


Figure 3. Local catchments (designated runoff areas 1-6) within the city center with historic core and indicated areas of retention potential. According to Greenery plan the EASTERN part of the city should be reviewed for the possibilities of local stormwater management. Own elaboration based on SCALGO Live Sweden.

Table 2. Summary table with an aerial distribution of each major land cover category for a baseline (m0) together with a distribution of land cover types and main groups (positive extremes are in bold-italic, negative in bold). The smallest catchments (No. 2 and 4) cover areas of 12.00 ha each, while the largest (No. 6) – 202 ha. Own elaboration based on SCALGO Live Sweden.

Land use/cover [category/ type]	Catchments						Area cumulative	
	Cat_1 [ha]	Cat_2 [ha]	Cat_3 [ha]	Cat_4 [ha]	Cat_5 [ha]	Cat_6 [ha]	[ha]	[%]
Mixed use	✓	✗	✗	✗	✗	✗		
Industrial	✓	✗	✓	✗	✓	✗		
Commercial	✓	✓	✓	✓	✓	✓		
Dense residential	✓	✓	✓	✓	✓	✓		
Low residential	✓	✗	✗	✗	✓	✓		
Upstream area	31.00	12.00	39.00	12.00	113.00	202.00	409.00	100
Artificial Cumulative	24.00	9.34	28.00	8.66	57.00	94.00	221.00	54.03
Natural Cumulative	6.24	2.54	11.00	3.37	56.00	108.00	187.15	45.76
Water Cumulative	0.00	0.00	0.00	0.00	0.27	0.00	0.27	0.07
Building	8.33	2.96	7.07	2.22	13.93	26.00	60.51	14.79
Paved road	4.79	1.75	3.97	1.70	8.03	26.00	3.24	10.57
Unpaved road	0.00	0.00	0.00	0.00	0.25	0.18	0.43	0.11
Other paved	11.64	4.63	17.00	4.75	34.00	45.00	117.02	28.61
Dense vegetation	3.86	1.53	5.13	2.25	20.87	85.00	123.83	30.28
Shallow vegetation	1.61	0.90	2.44	1.13	9.79	22.00	37.87	9.26
Bare land	0.70	0.03	2.94	0.01	26.06	1.62	31.26	7.67

3.2 Phase two: modeling (low-impact development, bio-density and urban greenery)

Results of the first part of analyses confirmed the formation of flow accumulation in the city's eastern part. Simulation of land cover, conducted in SCALGO Live Sweden with disabled option for the outflow into sewerage system, gave the results for Low-Impact Development (m1), Bio-density (m2) and Urban Greenery (m3). Each modeling focused on different types of land cover (see SF3. and SF4. and Tab. 3-4) of a fixed set of six hydrologically connected catchments. No inferential statistical tests were applied to compare catchments themselves.

An overall proportion of greenery has changed considerably. In LID modeling, the number of impervious spaces decreased the most in catchment No. 3 and 5 (industrial areas), the least in No. 6 (residential area with semi-detached houses). Bio-density modeling involving an introduction of multi-story vegetation (up to *Dense vegetation*) did not translate into fluctuations in principal land cover categories balance within catchments (focusing on enriching greenery structure). A proportion of artificial to natural areas was maintained. The most significant percentual increase in *Dense vegetation* type occurred in catchment No. 6 (dominant low residential, a change of almost 11%) the least – in No. 1 (dominant mixed-use, a change of 5%). In Urban Greenery, the most significant increase in *Dense vegetation* type occurred in catchment No. 5 (dominant industry, a change of almost 62%) and No. 3 (dominant industry, a change of over 57%), the least in No. 6 (dominant low residential; 34%). A proportion between natural and artificial areas has also changed remarkably. The most significant percentual increase in green spaces has occurred in catchment No. 3 (dominant industry; 45%), the least in No. 6 (dominant low residential; 22%).

SCALGO allows also for the determination of the amount of water accumulated in depressions (topographic characteristics) and the absorption capacity of soils (infiltrated or drained) (SF5. in supplementary materials). As shown in Tab. 3 and 4, scenario m3 resulted in the largest reduction in depression storage (DEP_m3) and the highest increase in infil-

trated/drained water (INF_m3) at the system level. For catchments 2 and 3 we obtained results showing increases in infiltration of up to 203% and 225% respectively. This occurred in areas with a very high proportion of artificial areas, which were replaced in the modeling by land cover consisting of dense vegetation. The quantitative change of LID (36% of overall area) involved a shift from artificial areas (*Bare Land* and *Other Paved*) to *Shallow vegetation*, amounting to 95% and 129% respectively, and the m2 simulation (Bio-density, 9% of overall area), the change involved a shift from *Shallow vegetation* to *Dense vegetation*, amounting to a maximum of 21% and 13% accordingly.

The consistency of results across catchments indicates that the conclusions are not sensitive to the choice of aggregation method. The consistency analysis was used to assess whether each scenario produced improvements in the same direction across catchments, defined as a reduction in depression storage ($p = 0.006$) and an increase in infiltrated/drained water ($p = 0.002$) relative to the baseline. The scatter plot (SF6.) revealed a clear differentiation between scenarios in terms of their joint, described above, effects. Scenario m3 consistently occupied the upper-left region of the plot, indicating simultaneous reductions in depression storage and increases in infiltrated/drained water across all catchments. This pattern suggests a robust and synergistic effect without observable trade-offs between the two indicators. In contrast, scenario m1 showed moderate improvements with greater variability among catchments, while scenario m2 exhibited only minor changes, with values clustered near zero. The results of Friedman's test suggest that differences between scenarios are systematic rather than random. However, given the small number of catchments and the deterministic nature of the modeling framework, the statistical analysis was interpreted as supportive evidence only.

As expected, each of the models improves the situation and is statistically significant. It is possible to quantify the impact of specific measures utilizing NBS on runoff volume, depending on the location and characteristics of the urban area. Pairwise, post-hoc comparisons did not reveal statistically signifi-

cant differences between individual scenarios after adjustment for multiple testing. Overall, scenario m3 consistently outperformed the other scenarios across all analytical levels, including individual catchments, system-level aggregation, and joint indicator response.

Table 3. Baseline and scenario-specific values of depression storage and infiltrated/drained water without the usage of sewerage system through the catchments and modeling. Own elaboration based on SCALGO Live Sweden.

Catchment	Base m0 [m³]		m1 [m³]		m2 [m³]		m3 [m³]	
	DEP_base	INF_base	DEP_m1	INF_m1	DEP_m2	INF_m2	DEP_m3	INF_m3
1	30 053	4 135	23 335	7 502	29 993	4 620	16 117	9 866
2	8 272	1 413	8 272	2 750	8 272	1 716	7 371	4 283
3	26 314	6 130	24 827	14 020	26 314	6 942	7 044	19 912
4	12 381	1 852	9 291	3 351	11 756	2 175	6 637	5 140
5	45 298	27 539	45 450	39 660	45 433	29 935	41 623	54 347
6	122 075	80 000	103 631	96 339	113 799	86 990	80 670	114 451
total	244 393	121 069	214 806	163 622	235 567	132 378	159 463	207 998
Δ total [m³]			-29 587	42 553	-8 826	11 309	-84 930	86 929
Δ total [%]			-12.1	35.1	-3.6	9.3	-34.8	71.8

Table 4. Changes in depression storage and infiltrated/drained water relative to baseline conditions in each scenario and catchment. Rates are presented in numerical and percentage terms. The nature of catchments determined by a degree of urbanization: 1,2,4 – inner city, historical; 6 – suburban; 3,5 – industrial. Own elaboration based on SCALGO Live Sweden.

Catchment	Area modeled/ altered [%]	SCENARIO	ΔDEP [m³]	ΔDEP [%]	ΔINF [m³]	ΔINF [%]
1	39.81	m1/LID	-6 718	-22.4	3 367	81.4
2	38.83	m1	0	0.0	1 337	94.6
3	51.13	m1	-1 487	-5.7	7 890	128.7
4	39.66	m1	-3 090	-25.0	1 499	80.9
5	53.15	m1	152	0.3	12 121	44.0
6	23.08	m1	-18 444	-15.1	16 339	20.4
Number of catchments with improvement				4		6
1	5.19	m2/BD	-60	-0.2	485	11.7
2	7.50	m2	0	0.0	303	21.4
3	6.26	m2	0	0.0	812	13.2
4	9.42	m2	-625	-5.0	323	17.4
5	8.66	m2	135	0.3	2 396	8.7
6	10.89	m2	-8 276	-6.8	6 990	8.7
Number of catchments with improvement				3		6
1	45.00	m3/UG	-13 936	-46.4	5 731	138.6
2	45.58	m3	-901	-10.9	2 870	203.1
3	57.39	m3	-19 270	-73.2	13 782	224.8
4	48.08	m3	-5 744	-46.4	3 288	177.5
5	61.91	m3	-3 675	-8.1	26 808	97.3
6	34.25	m3	-41 405	-33.9	34 451	43.1
Number of catchments with improvement				6		6

4 Discussion

Precipitation data from Katrineholm confirms that changes in the city's precipitation patterns are forecast (IPCC, 2019; IPCC, 2023; Muller et al., 2015), and periods of sudden and intense rainfall will alternate with periods of drought. This will lead to an increase in the negative effects of climate change in terms of excessive surface runoff. This also indicates that trends in the amount and temporal distribution of precipitation are consistent with the trends described for Northern Europe.

Blue-green Infrastructure is not being incorporated into planning in small towns, and even when it is (as in the strategy for Katrineholm), it does not fully address the water issue. The results for the first part of the research show that the problem of surface runoff has not been addressed in any of the planning documents that make up the city's existing adaptation strategy. However, thanks to the collection of their provisions concerning pro-environmental and pro-climate elements, it was possible to outline a preliminary direction towards resiliency in the scope mentioned and to perform simulations to verify their effectiveness. At least two accumulation sites adjacent to the conservation area have been identified in the eastern part, which confirms the prescriptions of the Greenery plan (Grönplan, 2018) to be validated. They also provoke the opportunity to retain the excess rainwater locally in order to minimize potential damage, delay surface runoff by infiltrating and storing more water *in situ*, and protect the most valuable areas within the city core from surface waters coming towards. In this way, protection can also extend to sensitive areas, such as historic sites (Staccione et al., 2024; Wang et al., 2024). In this context, our study confirmed that the application of NBS in historic areas of small and medium-sized town centers in a way that would reduce surface runoff and improve infiltration capacity can be possible. Furthermore, the hypothesis that even the permeabilization of impervious surfaces in the immediate city center can yield tangible benefits in terms of reducing surface runoff and increasing infiltration has been confirmed. Because the threat of flooding radiates mainly from neighboring sites, NBS can only counteract the excess surface runoff to a

reasonable level in the study area, even if minimal retention areas accompany the higher density found in urban centers. It has also been validated that even if changes to land cover in the immediate city center prove limited for various reasons, changes to land cover aimed at retaining water in adjacent areas may be conceivable. Hence, the objectives of the study have been achieved.

The results of our research confirm the hypothesis that NBS can serve as an adaptation measure to climate change by increasing infiltration capacity and thus reducing surface runoff (Paggioli et al., 2024; Staccione et al., 2024). This also demonstrates that climate-friendly measures to manage excessive surface runoff can be proposed and implemented through changes in land cover, also when considered on a catchment level (Iwanicki, 2007; Banasik et al., 2014; Sveriges Riksdag, 2018).

The results in land cover type transformations showcase how a qualitative (m2, Bio-density), quantitative (m1, LID) and qualitative-quantitative (m3, Urban Greenery) transfiguration can influence the distribution of sealed, pervious and green areas and so shape the image of an urban space. Research into stormwater management from a LID perspective (the key element of interest to us in this method is permeable pavements – PP) primarily concerns the reduction of surface runoff. We were unable to find any studies that provided results on changes in the infiltration conditions of sites where PP have been used. Nevertheless, a reduction in runoff volume was observed in Javan et al. (2025), reaching 17% (PP combined with rain barrels in Tehran, Iran); 18% in the study by Soltaninia & Eskandaripour (2026), also in Tehran; around 21% in the study in India (Sehrawat & Shekhar, 2025); and 3%-47% in the City of Normal-Sugar Creek Watershed, Central Illinois (Ahiablame & Shakya, 2016). A reduction up to 20% of surface runoff can be characterized as a meaningful improvement. In studies conducted in Switzerland (Cavadini et al., 2024), in every scenario involving the use of permeable pavements (using PP alone and in combination with other types of LID solutions), the implemented measure reduced runoff entering the urban drainage system by approximately 20%. In the research of Ruiz-Perez et al. (2022) the increase in infiltration for green spaces, combined with the use of

SUDS measures, was 72%, reducing freshwater consumption to 50%. Our aggregate result for all catchments studied is a 35% increase in infiltration, which can be seen as a substantial improvement, as well as 12% decrease in depression storage. The number of catchment areas in which both values have changed is 4 (out of 6).

Bio-density can enhance soil infiltration and reduce surface runoff indirectly by increasing vegetation cover, root density, organic matter inputs, and soil structural stability. As a metric describing the density and spatial distribution of vegetation or green infrastructure within an urban area, its effectiveness depends on how the method is applied and the characteristics it measures. In our study, Bio-density involved a land cover simulation from *Shallow Vegetation* to *Dense Vegetation*, but without the introduction of rain gardens, retention basins planted with vegetation and shrubs, or other features. Nor have any concurrent, long-term field studies been carried out that would take into account the contribution of evapotranspiration, improvements in soil saturation, vegetation maturity and health, biodiversity and other factors. The change in land cover concerned only 9% of the total study area, representing surfaces scattered and sprawled across the catchments (Yu et al., 2024), also being surrounded by compacted surfaces. Depression storage has decreased by 3.6% and infiltration increased by 9.3%. We observed a change in both values in only 3 catchment areas (out of 6).

The use of UG has been associated with the application of green infrastructure to achieve a cooling effect and to improve the thermal quality of urban spaces and ecosystem services. The use of Urban Greenery in the study by Uribe et al. (2022) in Costa Rica, in the context of reducing surface runoff, proved most effective in urbanized and residential areas along streets, with the proviso that permeable pavements are identified as the representative for this system/method, also as the best one. The extent to which urban greenery contributes to changes in the water absorption capacity of urban areas, particularly in relation to urban green spaces, is less well recognized in the literature. Our results, showing an increase in infiltration of almost 72%, indicate highly favorable conditions and a strong hydrological

effect, likely resulting from substantial changes in soil properties through land cover.

In our view, our results showcase a research direction that would require a broader and more comprehensive investigation, including a greater number of variants incorporating blue infrastructure elements and for more interconnected areas.

Our land cover studies are strongly determined by the classification of land cover types specified in the SCALGO program (defined by a *Curve number Cn* and soil capacity), which can be unique for the country in which the research is conducted. Software offers the advantage that other parameters closely linked to the area's water management (e.g. soil types, topography, land use) are predefined through government documents and satellite data anchored in the program. Furthermore, it must be emphasized that the effectiveness of measures depends heavily on the existing situation regarding land cover and land use – that is, the initial conditions – which calls for the implementation of smarter stormwater management systems at the planning level. Eventually, different remedial measures will be applicable in urban centers and in suburban areas.

The research carried out is based on a single site and the unique nature of the selected catchment types, which constitutes a limitation of the study. It should therefore be regarded as a test, pilot and illustrative study for interdisciplinary research at the intersection of spatial planning, landscape architecture and hydrology. The level of detail serves to provide a preliminary assessment of the occurrence of problem areas and their scale, as well as to evaluate the overall effectiveness of individual green-water planning systems. Our aim was to develop a preliminary surface runoff assessment that could be incorporated at an early planning stage, used as a resource in urban planning by policymakers, and serve as an integral part of climate change adaptation strategies for smaller towns. It may also prove equally useful in other suburban or inner-city areas of an industrial or residential nature, as well as in areas requiring regeneration, remediation, denitrification, or simply better adaptation to adverse climate change towards resiliency.

The authors of this study found no relevant literature – neither concerning areas outside Europe, nor, a fortiori, areas within the European Union or the European continent – devoted to the application of LID, UG and Bio-density in urban green spaces in small and medium-sized towns, with the aim of increasing infiltration whilst simultaneously reducing surface runoff in the modeling of land cover changes. The use of permeable pavements is highlighted in the literature for LID and UG as a representative solution and, at the same time, the best option for each of the methods studied.

5 Conclusions

A locally adapted strategy based on the specific characteristics of the city and region must be mandatory. Hence, one of the leading solutions should be to improve water retention as an affordable and achievable NBS. The effectiveness of these remedies should be tested regarding the amount of water retained. Moreover, we argue that the implementation of leading NBS (here tested: Low Impact Development, Bio-density, Urban Greenery) in landscape and urban planning should correspond to the scale of action (state administration level). It should be noted that both the problem of excessive surface runoff and the regional approach for possible preventive and remedial actions have not been noticed and addressed in the strategy of the case study city, Katrineholm.

Modeling 3 of Urban Greenery proved to be the most effective in terms of infiltration improvement in the research conducted. These studies can be used to identify, at an initial stage, the scale and nature of problems relating to water retention and possible remedies. This is, of course, closely linked to the quality and type of soil in the city and their water retention capacity, as well as to the topography. The preliminary results obtained in this way can be expanded upon through further, more in-depth research, and can be tentatively translated into urban planning solutions and blue-green infrastructure measures, which could be incorporated into

subsequent stages of local spatial planning schemes or urban development strategies in the form of guidelines and recommendations. The issues and problem areas identified in this way may also facilitate the work of designers and contribute to raising public and design awareness.

An enrichment of climate change adaptation strategies with guidelines for NBS in medium-sized urban areas can also occur in a form of minimum percentages of individual land cover types, where such implementation is possible, not only for new developments. Guidance on a minimum use of particular tools: unsealing the surface of sealed pavements or car parks, enriching biodiversity and bio-density of urban greenery by creating flower meadows or multi-story vegetation could contribute to enhancing the absorptive capacity of urban areas, also in central/urban foundation parts of cities, and reducing the runoff.

By enhancing the absorption capacity of the soil, it was possible to reduce the extent of the catchment areas adjacent to the historic center, thereby potentially protecting the historic part of the city center from the adverse effects of possible excess surface water runoff from neighboring districts. Such measures could contribute to a gradual increase in resilience to flash floods caused by climate change.

The methodology used in the following study was developed with smaller towns in mind equally affected by the negative consequences of climate change, which 1) do not have a climate change adaptation strategy yet, or 2) they do not see the need for one, and/or 3) they lack the financial resources to develop one, or all three of these to some extent. It may prove equally useful in other areas of larger cities or their suburban, inner-city, industrial or residential districts, as well as in areas requiring various urban-landscaping transformations, such as regeneration, reclamation, de-urbanization or simply better adaptation to adverse climate change impacts. Moreover, the topics in this area may serve as a research context for further scientific analysis.

The first area for implementing the general guidelines is the planning sector, and it may be used for

the strategic planning of (smaller) urban units. At the same time, this research may point the way to further work for specialists in many other fields, enabling in-depth research tasks, and could be replicated for a larger sample with similar or varied topographical and hydrological conditions.

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Declaration of competing interest

The authors declare not to have any known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to the use of contract-limited data from Lantmäteriet (Ref. no. LM2023/031735, (c) Lantmäteriet), but are available from the corresponding author on reasonable request.

Author contributions: CRediT

Barbara Warzecha: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Visualization, Writing – original draft, review and editing

Joanna Dudek-Klimiuk: Conceptualization, Funding acquisition, Methodology, Writing – review and editing

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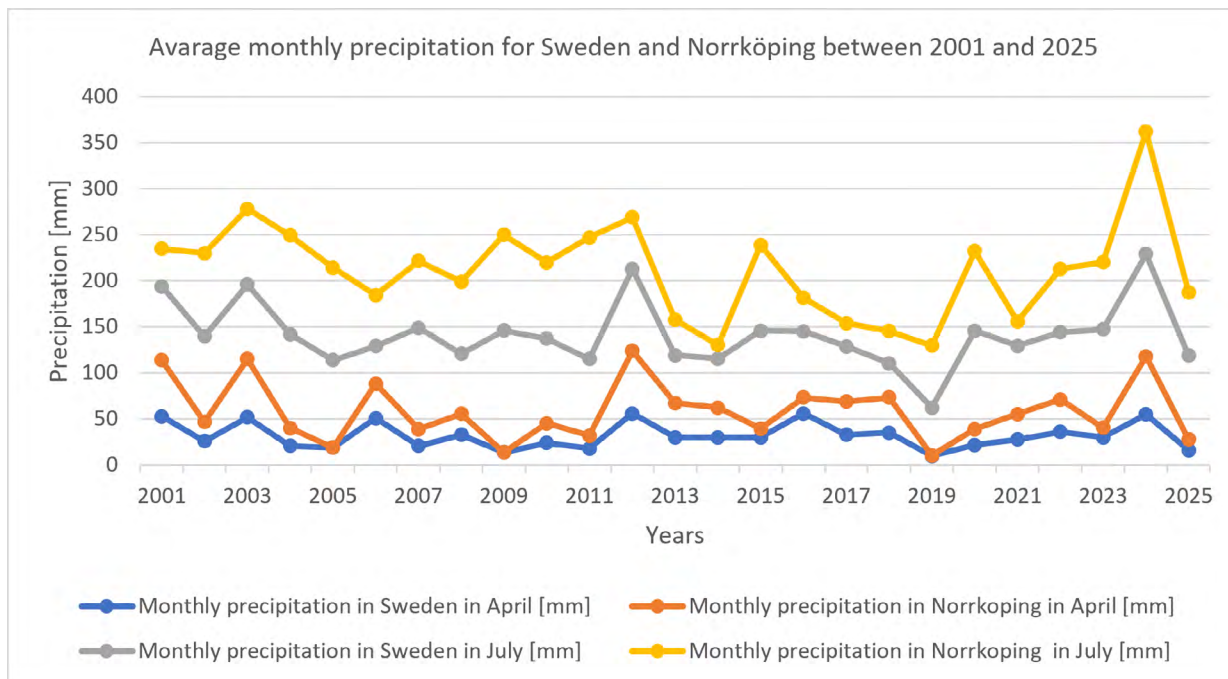
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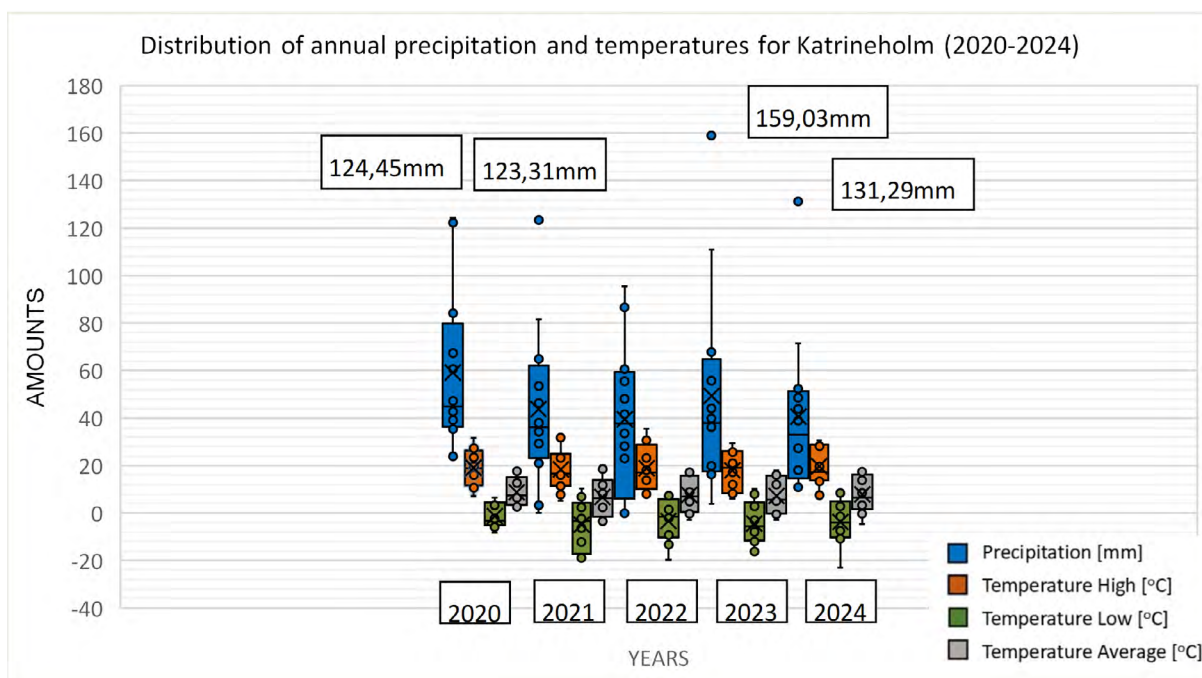
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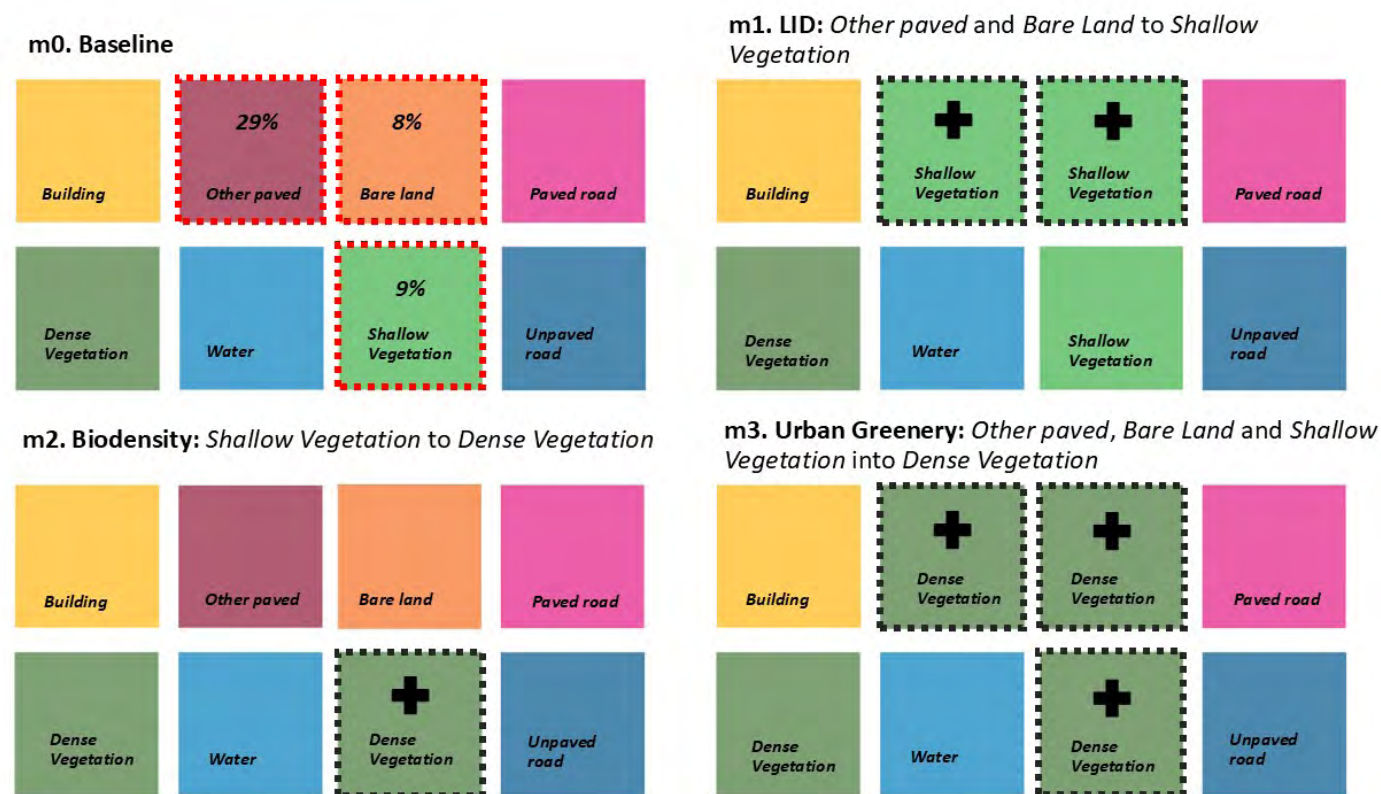
Supplementary Materials



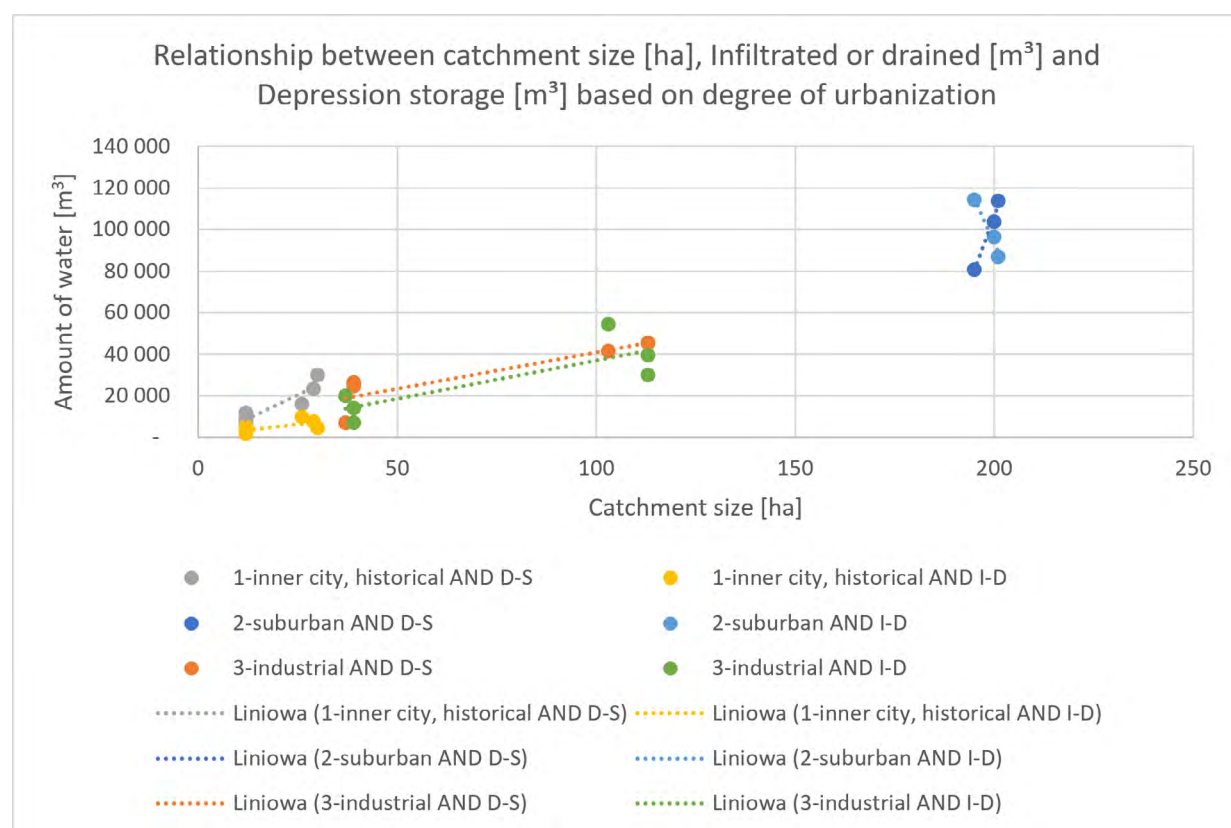
Supplementary Figure 1. Distribution of monthly average precipitation.



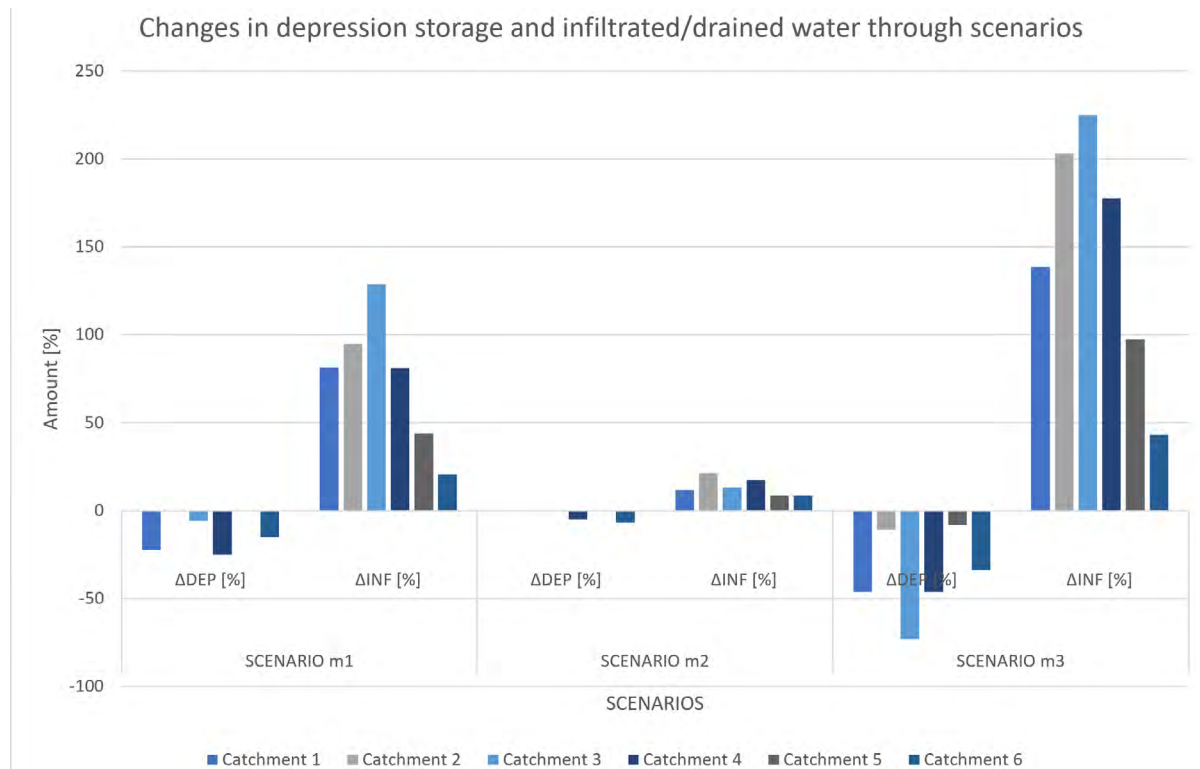
Supplementary Figure 2. Distribution of annual precipitation.



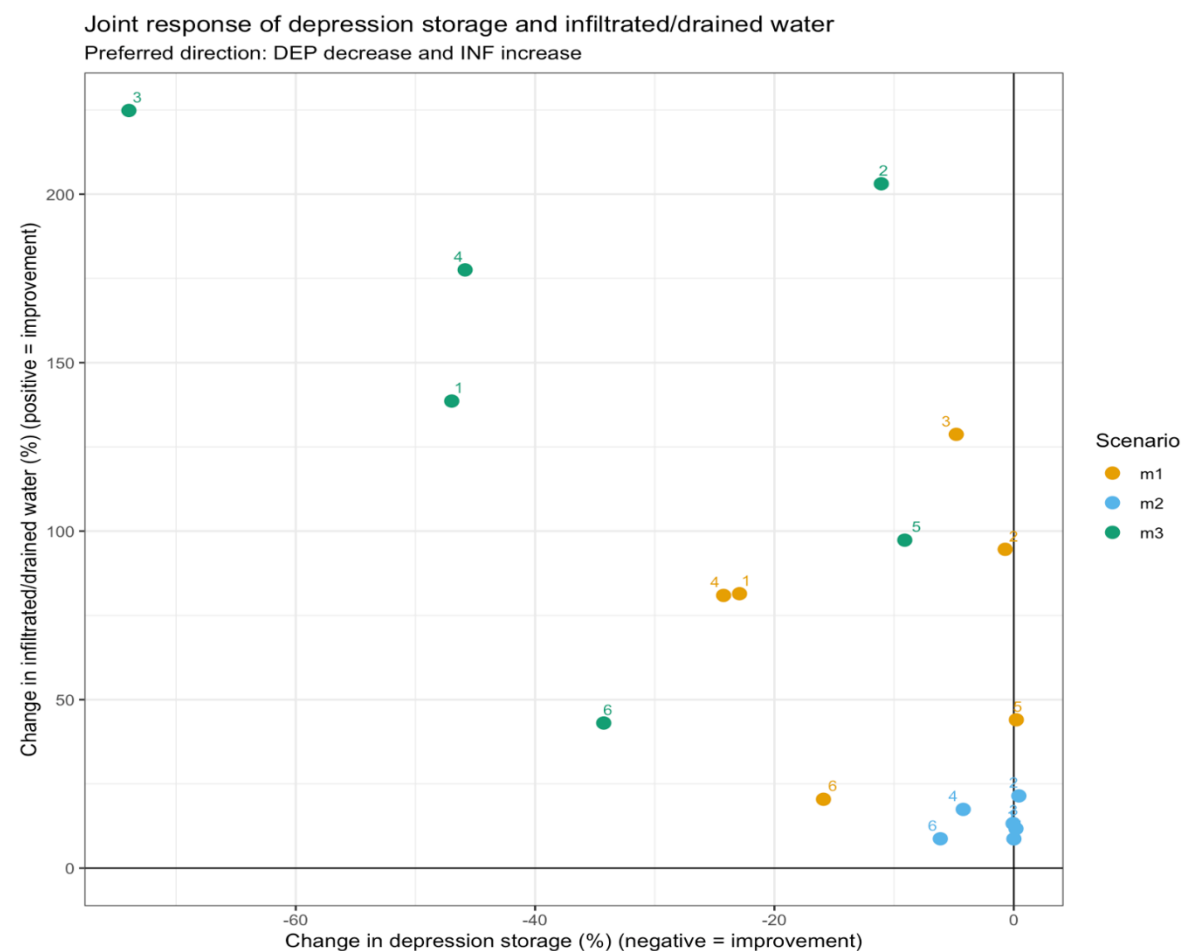
Supplementary Figure 3. Types of land cover subjected to change in each simulation.



Supplementary Figure 4. Relationship between catchment size, infiltrated or drained volume, and depression storage in modelings.



Supplementary Figure 5. Changes in depression storage and infiltrated-drained water through scenarios.



Supplementary Figure 6. Joint response of depression storage and infiltrated-drained water.

Supplementary Table 1. A summary of individual catchment areas by surface area, predominant land cover type, and the volume of depression storage and infiltrated or drained water. Scenarios: m0-baseline, m0_s-baseline with sewage, m1-LID, m2-Bio-density, m3-UG; nature of the catchment 1-inner city, historical (grey), 2-suburban (green), 3-industrial (yellow). Own elaboration based on SCALGO Live Sweden.

Catchment	SCENARIO	Predominant land cover type	Predominant land cover group (natural/artificial)	Degree of urbanisation	Area [ha]	Depression storage [m3]	Infiltrated or drained [m3]
1	m0	other paved		1	31	30 053	4 135
2	m0	other paved		1	12	8 272	1 413
3	m0	other paved		3	39	26 314	6 130
4	m0	other paved		1	12	12 381	1 852
5	m0	other paved		3	113	45 298	27 539
6	m0	dense vegetation		2	202	122 075	80 000
Total	m0	dense vegetation	artificial		409	244 393	121 069
1	m0_s	other paved		1	26	10 951	14 743
2	m0_s	other paved		1	12	5 279	6 599
3	m0_s	other paved		3	39	17 085	21 867
4	m0_s	other paved		1	12	5 141	6 408
5	m0_s	other paved		3	105	40 664	54 055
6	m0_s	dense vegetation		2	197	68 440	128 896
Total	m0_s	dense vegetation	artificial		386	147 560	232 568
1	m1	shallow vegetation		1	29	23 335	7 502
2	m1	shallow vegetation		1	12	8 272	2 750
3	m1	shallow vegetation		3	39	24 827	14 020
4	m1	shallow vegetation		1	12	9 291	3 351
5	m1	shallow vegetation		3	113	45 450	39 660
6	m1	shallow vegetation		2	200	103 631	96 339
Total	m1	shallow vegetation	natural		405	214 806	163 622
1	m2	other paved		1	30	29 993	4 620
2	m2	other paved		1	12	8 272	1 716
3	m2	other paved		3	39	26 314	6 942
4	m2	other paved		1	12	11 756	2 175
5	m2	other paved		3	113	45 433	29 935
6	m2	dense vegetation		2	201	113 799	86 990
Total	m2	dense vegetation	artificial		407	235 567	132 378
1	m3	dense vegetation		1	26	16 117	9 866
2	m3	dense vegetation		1	12	7 371	4 283
3	m3	dense vegetation		3	37	7 044	19 912
4	m3	dense vegetation		1	12	6 637	5 140
5	m3	dense vegetation		3	103	41 623	54 347
6	m3	dense vegetation		2	195	80 670	114 451
Total	m3	dense vegetation	natural		385	159 463	207 998